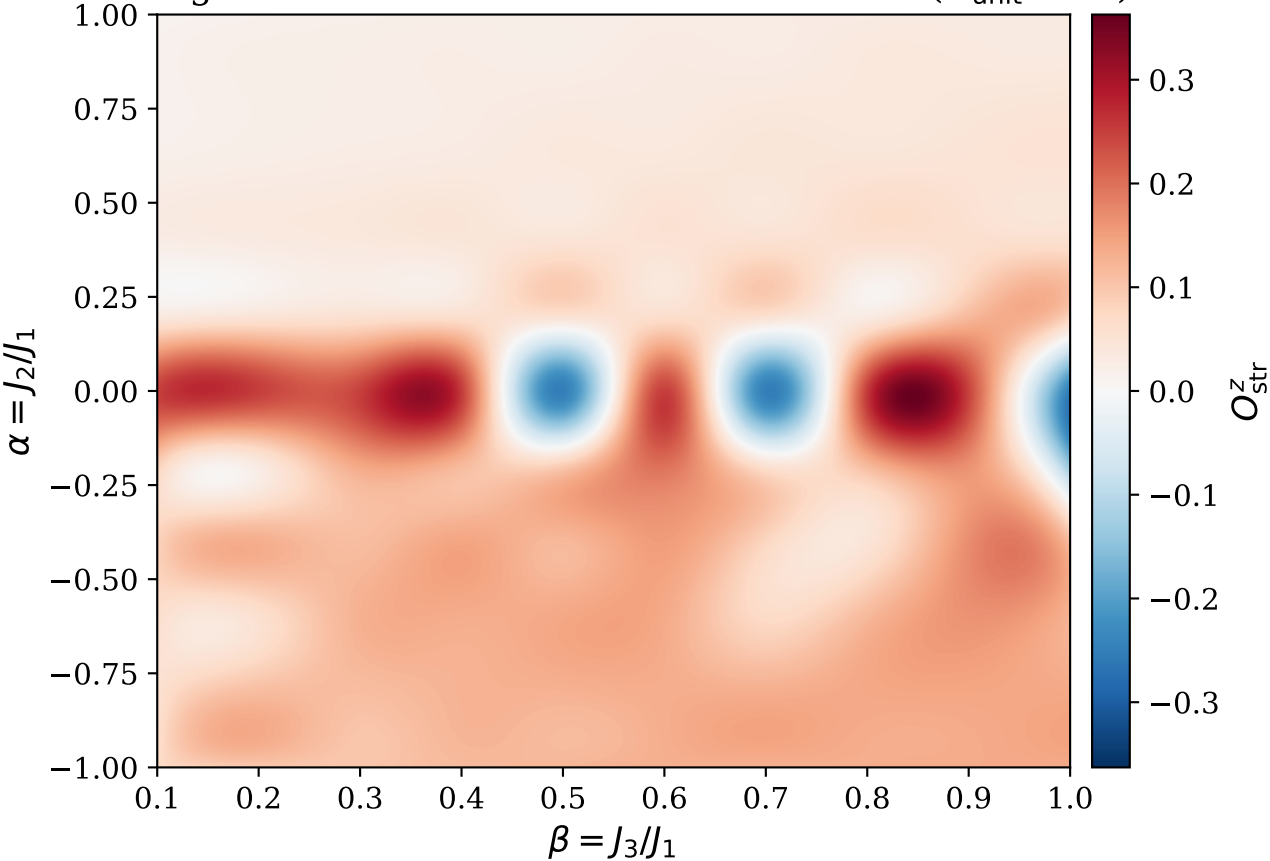

String Order Parameter — Pentamer Chain ($N_{\mathrm{unit}} = 12$)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
$\alpha = J_2/J_1$
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
$\beta = J_3/J_1$
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
O^z_{str}